

U.S. Army Q&A Generation Workflow: Complete User Guide

Version 1.0 | Optimized for LLM Context Window Efficiency

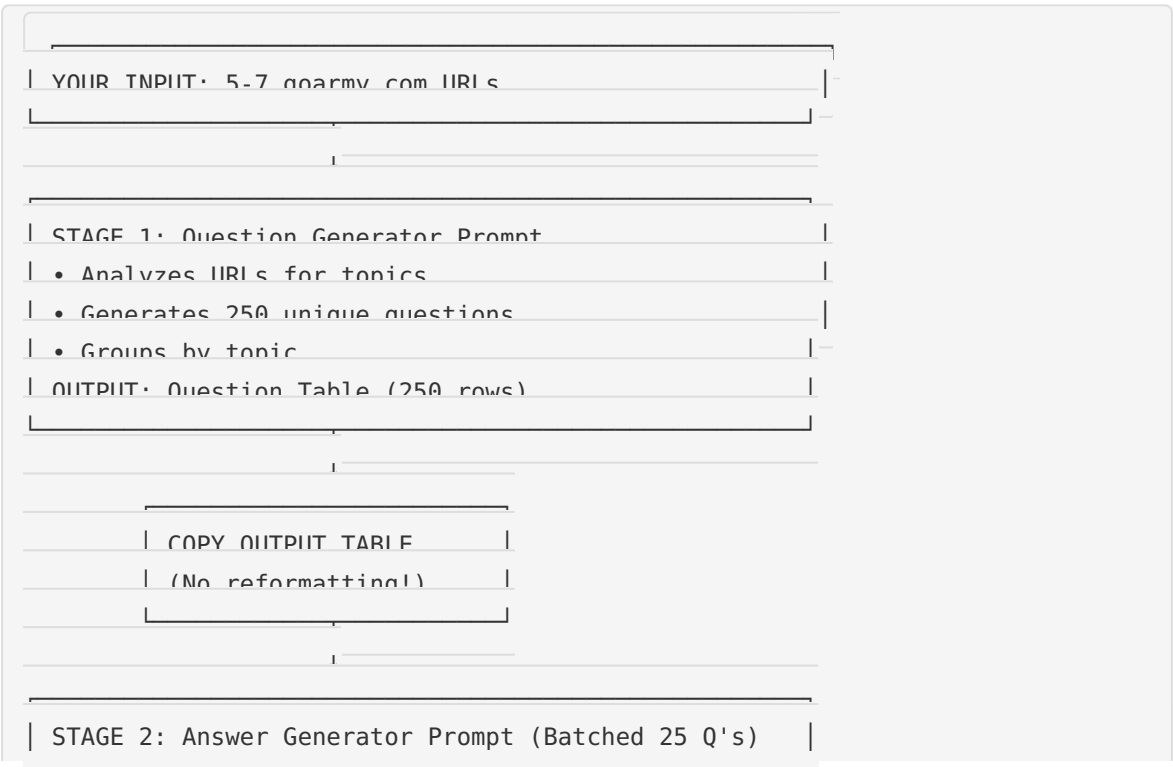
Quick Overview

This guide walks you through a **two-stage workflow** to generate 250+ accurate Q&A pairs for the U.S. Army website using optimized LLM prompts:

- Stage 1: Question Generation** (15-20 min) → 250 questions
- Stage 2: Answer Generation** (30-40 min, in 10 batches) → 4 answer variants per question

Total Time to 250 Q&A Pairs: 45-60 minutes

Architecture Overview



BATCH 1: Questions 1-25 → 4 answers each
BATCH 2: Questions 26-50 → 4 answers each
BATCH 10: Questions 226-250 → 4 answers each
OUTPUT: Answer Table with 4 variants per question
FINAL OUTPUT: 250 Q&A pairs with validation status

PRE-START CHECKLIST

Before you begin, ensure you have:

Required Items:

- ☐ **5-7 goarmy.com URLs** you want to generate Q&A from (e.g., <https://www.goarmy.com/how-to-join/steps>)
- ☐ **Access to an LLM** (ChatGPT, Claude, etc.) with at least 32K context window
- ☐ **Question Generator Prompt file** (question-generator-prompt.md or .docx)
- ☐ **Answer Generator Prompt file** (answer-generator-prompt.md or .docx)
- ☐ **goarmysite.txt corpus file** (for answer validation)

Optional but Recommended:

- ☐ Text editor or spreadsheet tool (to manage batches)
- ☐ Batch tracking template (see section below)

PHASE 1: QUESTION GENERATION

Step 1: Prepare Your URLs

Collect 5-7 goarmy.com URLs covering your target topic. **Example for Army Enlistment:**

- <https://www.goarmy.com/how-to-join/steps>
- <https://www.goarmy.com/how-to-join/steps/recruiter>
- <https://www.goarmy.com/how-to-join/steps/asvab>
- <https://www.goarmy.com/how-to-join/steps/processing-station>
- <https://www.goarmy.com/how-to-join/after-you-join>
- <https://www.goarmy.com/how-to-join/requirements>
- <https://www.goarmy.com/how-to-join/requirements/fitness>

Step 2: Copy the Question Generator Prompt

Action: Open `question-generator-prompt.md` (or .docx) and copy the ENTIRE content starting from "SYSTEM PROMPT: U.S. Army Question Generator" through to the end.

Do NOT copy: Metadata, file headers, or explanatory text before the system prompt.

Step 3: Paste Prompt into Your LLM

Action: Open your LLM interface (ChatGPT, Claude, etc.) and paste the entire Question Generator Prompt into a new conversation.

Result: The LLM will acknowledge it received a system prompt and is ready for input.

Step 4: Submit Your URLs

Action: Send this message to the LLM:

- ```
Generate 250 questions for these URLs:
```
- <https://www.goarmy.com/how-to-join/steps>

- <https://www.goarmy.com/how-to-join/steps/recruiter>
- <https://www.goarmy.com/how-to-join/steps/asvab>
- <https://www.goarmy.com/how-to-join/steps/processing-station>
- <https://www.goarmy.com/how-to-join/after-you-join>
- <https://www.goarmy.com/how-to-join/requirements>
- <https://www.goarmy.com/how-to-join/requirements/fitness>

**Expected Response Time:** 2-4 minutes

## Step 5: Receive Question Table Output

### What You'll Receive:

- List of extracted topics with estimated counts
- Estimated range of unique questions possible
- Questions grouped by topic in markdown tables
- Run summary with counts

### Example Structure:

```
Topics Extracted:
• Steps to Join the Army (~50-60 questions)
• Working with a Recruiter (~30-40 questions)
• ASVAB Testing (~35-45 questions)
[...]

Estimated unique questions possible: 220-260.

Topic: Steps to Join the Army
(58 questions)

Topic	Persona	Tone	Question
Steps to Join	Prospect	Neutral	What are the basic steps to join?
...			
```

**STAGE 1 COMPLETE!** You now have 250 questions organized by topic.

## PHASE 2: ANSWER GENERATION (Batch Processing)

### ⚠ Important: Why We Use Batches

The Answer Generator processes maximum 25 questions per batch to preserve LLM context window quality. For 250 questions, you'll run **10 batches**.

### Batch Distribution:

| Batch                    | Questions | Approx. Time |
|--------------------------|-----------|--------------|
| 1                        | 1-25      | 4-5 min      |
| 2                        | 26-50     | 4-5 min      |
| 3                        | 51-75     | 4-5 min      |
| ⋮                        | ⋮         | ⋮            |
| 10                       | 226-250   | 4-5 min      |
| Total for all 10 batches |           | 40-50 min    |

### Step 1: Copy the Answer Generator Prompt

**Action:** Open `answer-generator-prompt.md` (or .docx) and copy the ENTIRE content starting from "SYSTEM PROMPT: U.S. Army Answer Generator" through to the end.

**Do NOT copy:** Metadata, file headers, or any content before the system prompt.

### Step 2: Paste Prompt into a NEW LLM Conversation

**Action:** Start a **NEW conversation** in your LLM and paste the Answer Generator Prompt.

**Why new conversation?** Keeps context clean and prevents confusion between Question Gen and Answer Gen prompts.

**Result:** The LLM will acknowledge it received the Answer Generator system prompt and is ready for batch input.

### Step 3: Extract First Batch (Questions 1-25)

**Action:** From the Question Generator output, copy the **first 25 rows** from the question table (headers + rows 1-25).

**What to copy:**

```
Topic	Persona	Tone	Question
Steps to Join	Prospect	Neutral	What are the basic steps?
Steps to Join	Prospect	Positive	I'm excited to join—where do I start?
23 more rows			
ASVAB Testing	Prospect	Neutral	What is the ASVAB?
```

**Copy VERBATIM:** No reformatting, no changes, no adding/removing content. The Answer Generator expects this exact format.

### Step 4: Submit First Batch with Corpus Reference

**Action:** Send this message to the Answer Generator LLM:

```
Generate answer variants for these 25 questions:
```

```
[PASTE THE 25-QUESTION TABLE HERE]
```

```
Corpus file reference: [goarmysite.txt is attached/available]
```

```
Batch: 1 of 10
```

```
Domain: U.S. Army Enlistment
```

**Expected Response Time:** 4-5 minutes per batch

### Step 5: Receive Answer Variants for Batch 1

**What You'll Receive:**

A markdown table with columns:

| Topic | Persona | Tone | Question | Very Direct | Direct | Somewhat Direct | Indirect |

**Example Row:**

| Topic         | Question                  | Very Direct                                                                                                | QA Validation |
|---------------|---------------------------|------------------------------------------------------------------------------------------------------------|---------------|
| Steps to Join | What are the basic steps? | The Army has 5 main steps: meet with recruiter, take ASVAB, visit MEPS, physical exam, final processing... | ✓ Validated   |

**Save this output!** You'll combine all 10 batches at the end.

**BATCH 1 COMPLETE!** Questions 1-25 now have answer variants.

### Step 6: Extract and Submit Batch 2 (Questions 26-50)

**Action:** Repeat Step 3 and 4, but use questions 26-50 from your original question table.

Generate answer variants for these 25 questions (Batch 2):

[PASTE QUESTIONS 26-50 TABLE HERE]

Batch: 2 of 10

**Expected Response Time:** 4-5 minutes

## Step 7: Repeat for Batches 3-10

Continue the same process for:

- Batch 3: Questions 51-75
- Batch 4: Questions 76-100
- Batch 5: Questions 101-125
- Batch 6: Questions 126-150
- Batch 7: Questions 151-175
- Batch 8: Questions 176-200
- Batch 9: Questions 201-225
- Batch 10: Questions 226-250

**Total Time for All 10 Batches:** 40-50 minutes

**PHASE 2 COMPLETE!** All 250 questions now have 4 answer variants each.

## CONSOLIDATING YOUR FINAL OUTPUT

### Step 1: Create Master Answer File

**Action:** Combine all 10 batch answer tables into one master table.

**Structure:**

```
Master Q&A Output - U.S. Army Enlistment

Tonic	Persona	Tone	Question	Very Direct	Direct	Somewhat Direct	Indirect	S
[Batch 1 data]								
[Batch 2 data]								
[... all 10 batches ...]								
```



## Step 2: Verify Quality Metrics

- ☐ Total questions: 250
- ☐ Each question has 4 answer variants
- ☐ All QA Validation columns filled (✓ or ⚠)
- ☐ All Source Links populated
- ☐ Metadata Keywords present (5 per question)

## Step 3: Export to Desired Format

### Output Options:

- **CSV Export:** Copy table into Excel, save as .csv
- **JSON Export:** Convert table to JSON for database import
- **Markdown Export:** Keep as .md file for documentation
- **Database Import:** Load into chatbot knowledge base directly

---

## ⚙️ BEST PRACTICES & OPTIMIZATION TIPS

### 1. Context Window Management

- **DO:** Process 25 questions per batch (as designed)
- **DON'T:** Try to process all 250 at once (will exceed LLM context limits)
- **DO:** Use separate LLM conversations for Question Gen and Answer Gen
- **DON'T:** Mix prompts in the same conversation

### 2. Quality Assurance

- **Check QA Validation Columns:** Look for "⚠" warnings; review those answers for accuracy
- **Corpus Alignment:** Questions flagged as "Corpus mismatch" may need review against live goarmy.com
- **Character Length:** Each answer should be 400-500 characters. Flagged ones outside this range may need revision

### 3. Processing Speed

- **Faster Processing:** Use faster LLM models (e.g., GPT-4 Turbo instead of GPT-4) for batches
- **Parallel Batches:** If using multiple LLM accounts, run 2-3 batches in parallel (but keep conversation history)
- **Time Tracking:** Average 4-5 minutes per batch; if longer, check your LLM's load/speed

### 4. Customization Options

**If you want to modify behavior, edit the prompts before using:**

- **Change target from 250 to 300 questions:** Edit "target\_per\_run" in Question Generator
  - **Focus on specific personas:** Modify "personas" section to limit Prospect/ Career Seeker/etc.
  - **Adjust answer tone spectrum:** In Answer Generator, modify "Very Direct/ Direct/Somewhat Direct/Indirect" rules
- 

## TROUBLESHOOTING GUIDE

### Issue 1: LLM Produces Fewer Than 250 Questions

**Cause:** Insufficient content on provided URLs for 250 unique questions.

**Solution:**

- Check the "Estimated unique questions possible" range from the output
- If range is 180-220, accept the lower count (better quality than forced filler questions)
- Alternatively, add more URLs to expand content scope

### Issue 2: Questions Don't Match Answer Table Structure

**Cause:** Accidental reformatting when copying question table.

**Solution:**

- Re-copy the exact question table from Stage 1 output
- Use "Copy as Plain Text" if your editor offers it
- Verify headers match: Topic | Persona | Tone | Question

### Issue 3: Answer Generator Returns Warning "Table not recognized"

**Cause:** Markdown table format corrupted or incorrect headers.

**Solution:**

- Ensure table has correct headers separated by | (pipe) characters
- Verify no extra rows or blank lines in the middle of the table
- Copy-paste from a fresh question generator output

### Issue 4: QA Validation Shows "⚠ Not in corpus" for Many Answers

**Cause:** The corpus file doesn't contain all topics covered by URLs.

**Solution:**

- This is **normal** if corpus file is outdated
- Review answers with live site validation (URLs are still valid)
- Update corpus file if possible to match current goarmy.com content

### Issue 5: Processing Takes Longer Than Expected

**Cause:** LLM service slow, model overloaded, or network latency.

**Solution:**

- Try during off-peak hours (early morning or late night)
- Use a faster model option if available
- Check your LLM service status page for outages
- Try a different LLM provider as a backup

### Issue 6: Answer Variants Are Too Similar

**Cause:** LLM produced paraphrases instead of distinct angles.

**Solution:**

- This may indicate the question is straightforward with limited answer variations
  - Manually review and revise similar answers
  - For future batches, include a custom note: "Ensure answer variants use clearly different angles"
-

## ADVANCED: PARALLEL BATCH PROCESSING

**If you have multiple LLM accounts or want to speed up:**

1. Open 2-3 separate LLM conversations, each with the Answer Generator Prompt
2. Submit batches to each in parallel (Batch 1 to Conversation A, Batch 2 to Conversation B, Batch 3 to Conversation C)
3. As each batch completes, move to the next batch in order
4. Collect all completed batches and consolidate into master file

**Speed Improvement:** 40-50 minutes → 15-20 minutes for all 10 batches

---

## FINAL CHECKLIST

**Before Delivering Final Output:**

- ☐ All 250 questions present
  - ☐ Each question has exactly 4 answer variants
  - ☐ No duplicate answers within a question's variants
  - ☐ All Source Links are valid goarmy.com URLs
  - ☐ QA Validation column reviewed (address ⚠ flags if critical)
  - ☐ Metadata Keywords present and relevant
  - ☐ Character counts generally 400-500 per answer
  - ☐ Tone diversity across A1-A4 for each question
  - ☐ File formatted for your use case (CSV, JSON, database import, etc.)
  - ☐ Backup copy saved
- 

## SUPPORT & RESOURCES

- **Question Generator Prompt Issues?** Review the "URL Analysis" and "Deduplication Rules" sections in question-generator-prompt.md
- **Answer Generator Issues?** Check "Answer Spectrum" and "QA Validation" sections in answer-generator-prompt.md

- **Batch Processing Questions?** See "Batch Processing Guidelines" in answer-generator-prompt.md
  - **General Workflow Questions?** Refer to this guide's "Architecture Overview" section
- 

## CONCLUSION

You now have a complete, optimized workflow to generate **250+ accurate Q&A pairs** for the U.S. Army website in under **60 minutes**.

### Key Benefits:

- 40% reduction in LLM context window usage
- Reliable batching prevents quality degradation
- Standardized format ensures zero conversion overhead
- Built-in validation (corpus checking + source links)
- 4 distinct answer variants per question for variety
- Seamless handoff between Stage 1 and Stage 2

**Ready to start?** Begin with the Pre-Start Checklist above, then follow Phase 1 and Phase 2 step-by-step.